

Energy, Emissions and AI

Posted by u/cognitive_courier • • 0 points • 2 comments

Hey! I write a newsletter covering the AI and tech world, and just wrote an article about the intersection of energy, emissions and AI that I wanted to share with you.

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Artificial intelligence (AI) has the potential to revolutionize the energy sector, but it also brings significant challenges. As AI systems and data centers expand, their electricity consumption is straining power grids, potentially increasing greenhouse gas emissions. Currently, AI computing accounts for less than 1 percent of global emissions, but this could grow substantially depending on the pace of AI advancement and energy choices. Companies like Google aim for net-zero emissions by 2030, yet their emissions have surged, driven by AI and data center expansions.

In the medium term, AI can enhance energy production and efficiency, contributing to cleaner energy use. AI systems can optimize renewable energy generation, predict weather patterns, and manage energy storage, helping to balance supply and demand. Additionally, AI can accelerate research by analyzing vast datasets to identify promising materials for batteries, carbon capture, and other technologies. This could lead to significant advancements in energy production, such as breakthroughs in nuclear fusion and hydrogen energy, while reducing costs and emissions.

Looking ahead, AI's long-term impact on energy is uncertain but potentially transformative. Advanced AI systems could simulate entire societies to develop climate-friendly policies with broad acceptance. However, as AI grows more powerful, it will require significantly more energy, driving demand for cleaner and more efficient energy sources. The Gulf countries, with their resources and focus on AI, could benefit greatly by integrating AI with human expertise to address key challenges. Ensuring the availability of reliable and transparent data will be crucial for maximizing AI's potential in transforming the energy landscape.

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I would be curious to hear from others - what are some specific ways AI can optimize renewable energy generation and storage? Do you think tech firms such as Google will ever

achieve their emissions goals? What are the potential risks and benefits of AI systems simulating entire societies to develop climate-friendly policies?

If you enjoyed this you can see more of my work at '[The Cognitive Courier]' (<https://cognitivecourier.com>)'.

Comments

u/PanflightsGuy • 1 points • 2024-07-08 22:41 UTC

I asked Bing's Copilot to suggest websites that could help me plan my travels in a more sustainable fashion. I asked for real emission cuts, not carbon offsetting. And I was quite impressed.

It suggested a website that replaces indirect flights with direct flights plus train or bus combinations. Often those trains or buses connect at the airport itself, meaning you don't have to get to the city center to reach them. Such combos typically reduce the emissions quite a bit compared to the connected flight variants.

For instance, say you go from Oslo to Glasgow. There are no direct flights. But you can fly to Edinburgh or Manchester and take the train or bus from the airport to Glasgow city center.

You can even take the train from Oslo to Stavanger, fly to Aberdeen, and take the train to Glasgow from there. Emissions will with the last combo be about 65% lower than if flying connected via say Amsterdam.

So the AI's often know how to make emission reductions. The thing is, climate cutting sites often do not have massive marketing budgets. And without the marketing they usually won't have many backlinks or much traffic. Which is often needed to rank for popular transactional queries.

u/cognitive_courier • 1 points • 2024-07-10 06:10 UTC

Thank you for this - it's super interesting. I discuss use cases in my newsletter and I'm going to use this example this week, it's something I haven't even thought of.

u/humundo • 1 points • 2024-07-08 15:57 UTC

The biggest problem in energy right now is that the incumbent firms and utilities (who will never be replaced) have no interest in changing anything about the way they function. Unless you have an AI that can force these companies to embrace new tech in opposition to basically their entire cost structure, AI will not be able to make the impacts you have described.

u/cognitive_courier • 1 points • 2024-07-10 06:09 UTC

Although I agree a big problem is existing players wielding too much power with no desire for change, I still believe the tech can bring in new efficiencies.

Just because the BPs and Shells of the world are immoral and act only in their interests, I don't believe that is a reason to ignore a good thing.